

Pre-Survey Instrument

The Informed Consent Form (ICF) is designed to protect your rights as a participant and can be accessed through [IRB Institute Approved Anonymous Link]. Please take your time to read it carefully and ensure you fully understand the research purpose, objectives, and methods. Participation is completely voluntary, and you may choose not to participate or withdraw at any time without any penalty or consequences.

By confirming that you have reviewed the research details and understand its nature, objectives, and methods, you declare that you are over 18 years old and agree to participate by completing the form.

If you select Yes, you will proceed to a short pre-survey, which will take approximately 5-8 minutes to complete.

If you select No, the form will close, and your autonomy and privacy will be fully respected.

- ☐ Yes (1)
- ☐ No (2)

1. What is your current education level?

- ☐ Undergraduate Student (1)
- ☐ Master Student (2)
- ☐ PhD Student (3)
- ☐ Postdoc (4)
- ☐ Other (5)

2. Which year are you in?

- ☐ Year 1 (1)
- ☐ Year 2 (2)
- ☐ Year 3 (3)
- ☐ Year 4 (4)
- ☐ Year 5 (5)
- ☐ Other (6)

3. What gender (if any) do you identify yourself as (check all that apply)?

- ☐ Male (1)
- ☐ Female (2)
- ☐ Non-binary or Gender-diverse (3)
- ☐ Prefer not to say (4)
- ☐ Other (5)

4. How would you describe your racial or ethnic background?

- ☐ American Indian or Alaska Native (1)

- ☐ Asian (2)
- ☐ Black or African American (3)
- ☐ Hispanic or Latino (4)
- ☐ Native Hawaiian or Other Pacific Islander (5)
- ☐ White/Caucasian (6)
- ☐ Prefer not to say (7)
- ☐ Other (8)

5. How long is your programming experience?

- ☐ Less than 1 year (1)
- ☐ 1-2 years (2)
- ☐ 2-3 years (3)
- ☐ 3-4 years (4)
- ☐ 4-5 years (5)
- ☐ More than 5 years (6)

6. How many years of industry experience do you have? (Including internship)

- ☐ Less than 1 year (1)
- ☐ 1-2 years (2)
- ☐ 2-3 years (3)
- ☐ 3-4 years (4)
- ☐ 4-5 years (5)
- ☐ More than 5 years (6)

7. In general, how frequent do you use AI tools (e.g., coding assistants like GitHub Copilot, Codeium, or general AI tools like ChatGPT) to assist with your programming activities (e.g., code generation, debugging, or learning new concepts)?

- ☐ Never (1)
- ☐ I tried these tools but stopped using them (2)
- ☐ Rarely (monthly or less) (3)
- ☐ Sometimes (weekly or less) (4)
- ☐ Often (1-2 times daily or nearly daily) (5)
- ☐ Always (3 or more times daily) (6)

8. How long have you been using AI tools (e.g., GitHub Copilot, Codeium, or ChatGPT) to assist with your programming activities?

- ☐ I don't use AI tools (1)
- ☐ Less than 3 months (2)
- ☐ 3 to 6 months (3)
- ☐ 6 months to 1 year (4)
- ☐ 1-2 years (5)
- ☐ 2-3 years (6)
- ☐ More than 3 years (7)

9. How long have you been using a version control platform? (e.g., platforms like GitHub, GitLab, Bitbucket, or others that allow developers to store, manage, and collaborate on code using version control systems such as Git.)

- ☐ Less than 1 year (1)
- ☐ 1-2 years (2)
- ☐ 2-3 years (3)
- ☐ 3-4 years (4)
- ☐ 4-5 years (5)
- ☐ More than 5 years (6)
- ☐ I don't know about these version control platforms (7)
- ☐ I heard these before, but did not use these (8)
- ☐ I tried these before, but gave up (9)

10. How frequently do you use version control platforms for personal or professional projects?

- ☐ Rarely (monthly) (1)
- ☐ Sometimes (weekly) (2)
- ☐ Often (once daily) (3)
- ☐ Always (more than once daily) (4)
- ☐ I don't know about version control platforms. (5)
- ☐ I heard these before, but did not use these for personal or professional projects. (5)
- ☐ I have tried using version control platforms before, but gave up. (6)

11. Which of the following best describes your level of experience with REST APIs? (REST APIs are interfaces that allow applications to communicate and exchange data, commonly used for backend services in software development.)

- ☐ I have never used REST APIs before. (1)
- ☐ Beginner (I am learning how REST APIs work and have only practiced making basic requests, like GET or POST, in tutorials or small exercises.) (2)
- ☐ Intermediate (I can use REST APIs for common tasks, such as retrieving data from endpoints, and have implemented simple integrations in personal or small projects.) (3)
- ☐ Advanced (I have extensive experience working with REST APIs, including designing, testing, and integrating multiple endpoints across several projects.) (4)
- ☐ Expert (I am proficient in designing, deploying, and managing REST APIs in production environments, including handling authentication, rate limiting, and versioning.) (5)

12. How frequently have you used MySQL?

- ☐ Rarely (monthly) (1)
- ☐ Sometimes (weekly) (2)
- ☐ Often (once daily) (3)
- ☐ Always (more than once daily) (4)
- ☐ I don't know about MySQL. (5)

- o I have tried using MySQL before, but gave up. (6)

13. How familiar are you with writing and executing SQL queries?

- o I have never written a SQL query before. (1)
- o Beginner (I am learning basic SQL concepts and have written simple queries, such as SELECT or INSERT.) (2)
- o Intermediate (I can write moderately complex queries involving JOINS, WHERE clauses, and aggregations.) (3)
- o Advanced (I am comfortable writing and optimizing complex queries, including subqueries and stored procedures.) (4)
- o Expert (I regularly write advanced SQL queries, manage databases, and handle performance tuning in production environments.) (5)

14. How familiar are you with Java Database programming?

- o I have never worked with Java database programming. (1)
- o Beginner (I can establish basic database connections in Java and execute simple queries using JDBC or similar tools.) (2)
- o Intermediate (I can write moderately complex Java code for database operations, including using prepared statements, handling transactions, and debugging errors.) (3)
- o Advanced (I am comfortable working with connection pooling, optimizing database interactions in Java, and using frameworks like Hibernate or Spring Data.) (4)
- o Expert (I regularly design and implement database-heavy Java applications, optimize performance, and handle distributed transactions or complex ORM configurations.) (5)